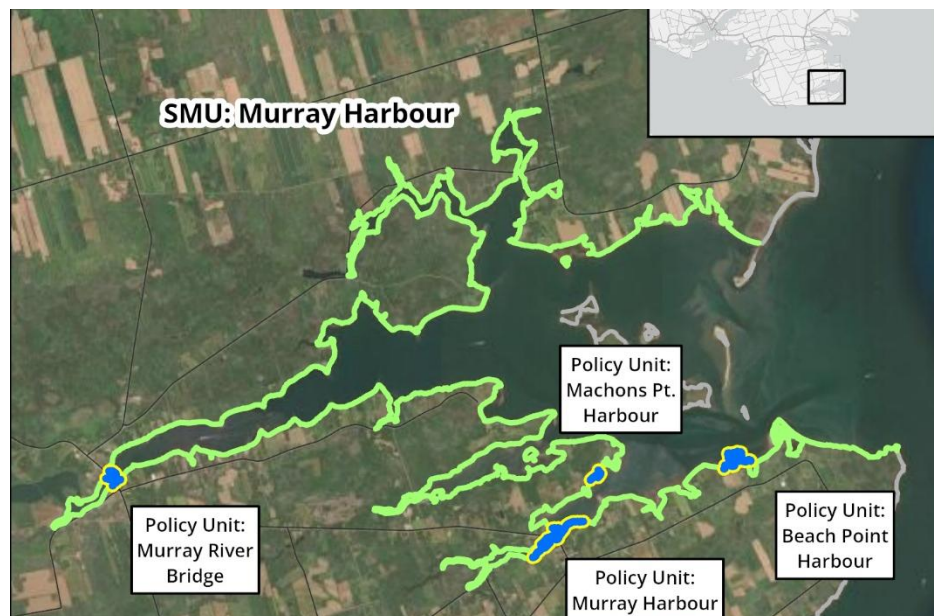


Murray Harbour

Shoreline length: 92 km



Shore Types

Murray Harbour contains multiple estuaries including one connected to the Murray River. Located on the east end of PEI, with a majority of low plains and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
64	5	4	2	21	2	2	0

Governance, Infrastructure, and Economy

The Murray Harbour SMU includes the rural municipalities of Murray Harbour and Murray River (municipalities with no planning, ~15% combined), with the remaining ~85% being unincorporated.

Coastal infrastructure includes active wharves and the roads leading to them, including Point Pleasant Road (Route 17); Girl Guide Camp Road (Murray River); the Point Pleasant Road Bridge (Little Basin); Route 17 (Cambridge); Mink River Road (Murray Harbour); and Beach and Poverty Roads (Murray Harbour North).

Other infrastructure includes the Beach Point Lighthouse; Wharf Lane, Beach Point, the Murray Harbour Range Rear Light; Keeping Road (Murray Harbour, at the beach); Murray Lane (Murray Harbour); Main Street and Riverside Drive; Westview Drive; the bridge and Confederation Trail crossing over the Murray River; the Murray River Sewage Lagoon; Machon Point Road; Fox River Road; Gladstone Road (Route 348, Murray Harbour); Route 4 Main Street (Murray River); the Murray River dam; Marina, Wharf at Big Basin, Murray Harbour North and Livingstone Road (Route 348, Murray River).

Coastal economic drivers include the Seal Cove Campground (Murray Harbour North), the Murray Harbour North wharf, Machon Point Road wharf and industry, the Murray Harbour wharf and marina, and the Beach Point Lobster Processing Company and wharf (Murray Harbour). Local and legacy land use constraints include several undeveloped small-lot subdivisions, with several lots (many undeveloped), and unpaved roads.

Natural Assets and Heritage

Coastal parks and cultural sites include Kings Castle Provincial Park, Murray River Memorial Park (playground), Railhead Park (playground), the Beach Point Lighthouse, and Black Brook Lookout Point (hiking/park).

The shoreline includes critical habitat for bird species including the endangered Piping Plover.

Planning and Policy Framework

Neither Murray Harbour or Murray River have official plans and provincial rules apply throughout this SMU. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Coastal Flood and Erosion Risk

Infrastructure along low plain, wetlands, and sand dune shorelines are vulnerable to increasing flood hazards. Erosion hazard is also highest along sand dunes and wetlands. Wetland migration should be allowed to follow rising sea levels to avoid coastal squeeze

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Murray Harbour				
Flood risk	61 [1/km]	74 [1/km]	88 [1/km]	111 [1/km]
Erosion risk	N/A	29 [<1/km]	38 [<1/km]	108 [1/km]
PU Murray Harbour				
Flood risk	9 [5/km]	14 [8/km]	19 [11/km]	28 [16/km]
Erosion risk	N/A	3 [2/km]	5 [3/km]	8 [5/km]
PU Beach Point Harbour				
Flood risk	8 [5/km]	8 [5/km]	8 [5/km]	10 [6/km]
Erosion risk	N/A	2 [1/km]	2 [1/km]	10 [6/km]
PU Murray River Bridge				
Flood risk	7 [12/km]	8 [13/km]	11 [18/km]	13 [22/km]
Erosion risk	N/A	2 [3/km]	2 [3/km]	15 [25/km]
PU Machons Pt Harbour				
Flood risk	4 [1<1/km]	6 [15/km]	7 [18/km]	9 [23/km]
Erosion risk	N/A	1 [3/km]	1 [3/km]	4 [1<1/km]

Sediment transport - The estimated net sediment transport rate averaged across the SMU is low. Sediment transport rate potential is highest along the exposed portion of Beach Point, which feeds sand formations at the entrance of the bay.

Proposed SMP Strategies

The primary strategy for Murray Harbour is NAI followed by MR for assets at risk. This is particularly important for outer areas supplying sediment to the beaches fronting the bay, such as Beach Point which protects the harbour. In areas where infrastructure is at risk, managed realignment and resist strategies are options.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	MR	MR
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets, while preserving natural shorelines with increasing flood hazard.		
Proposed PU strategy - Murray Harbour Waterfront	R	R	MR
Rationale	Resist strategy is warranted for critical infrastructure, but will not mitigate long-term flood risk. Long-term MR should consider raising land/buildings and/or retreat.		
Proposed PU strategy - Beach Point Harbour	R	MR	MR
Rationale	Resist strategy is warranted for critical harbour infrastructure, but will not mitigate long-term flood risk. Mid- to long-term MR should consider raising buildings and/or retreat.		
Proposed PU strategy - Murray River Bridge	R	R	MR
Rationale	Resist strategy is warranted for critical infrastructure, but will not mitigate long-term flood risk. Long-term MR should consider raising land and/or retreat.		
Proposed PU strategy - Machons Pt Harbour	R	MR	MR
Rationale	Resist strategy is warranted for critical harbour infrastructure, but will not mitigate long-term flood risk. Mid- to long-term MR should consider raising buildings and/or retreat.		



LEGEND

- Policy Unit
- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

0 325 650 1,300
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:

**COASTAL EROSION
AND FLOOD HAZARD**

**Shoreline Management Unit:
Murray Harbour**

Prince Edward Island
Île-du-Prince-Édouard

CBCL